

AI Invoice Reminder & Payment Tracker

End-to-end n8n pipeline: automated payment reminders, AI-drafted emails, and a payment webhook — technical & process documentation

AI Automation & n8n Workflow Developer

n8n

Groq LLM

Google Sheets

Email

This report documents an n8n automation that chases unpaid invoices without anyone having to remember to. Every day it scans a Google Sheets invoice ledger, works out exactly which clients need a nudge and how urgent that nudge should be, has Groq draft a professional reminder email for each one, and keeps the ledger and a finance team summary up to date. A separate webhook lets a payment page mark an invoice as paid the moment it's settled, stopping reminders immediately.

1. Executive Summary

Chasing unpaid invoices manually means someone has to remember who owes what, decide how firmly to word each follow-up, and keep track of who's already been reminded. This workflow automates all of that: a rule-based scheduler decides exactly when each client is due for a reminder and how urgent it should sound, Groq drafts the actual wording, and every send is logged so nobody gets double-reminded or missed. The moment a payment comes in through the webhook, that client drops out of the reminder cycle automatically.

Metric	Detail
Reminder trigger	Daily schedule (9:22 AM) reading every row of a Google Sheets Invoices tab
Reminder logic	Rule-based tiers: upcoming, due today, first/second reminder, final notice — by days overdue and days since the last reminder
Email drafting	A Groq-powered agent (gpt-oss-120b) writes the message; the workflow adds the invoice details table and payment button itself
Payment webhook	A separate endpoint marks an invoice paid, sends a confirmation email, and stops future reminders for it
Audit trail	Every reminder sent is appended to an Activity Log tab, in addition to updating the invoice row
Team visibility	A daily summary email totals reminders sent, amounts outstanding and urgency breakdown for the finance team

2. Business Value

- Nobody has to remember to chase anyone — the ledger is checked every single day, automatically, at the same time.
- Reminders get firmer as they need to — a gentle heads-up before the due date escalates to a final notice past 30 days, without a human deciding the tone each time.
- No double-reminding or awkward silence — each tier only fires once enough days have passed since the last reminder, so clients aren't nagged daily or forgotten for weeks.
- Payments stop reminders instantly — a payment page hitting the webhook marks the invoice paid and the client immediately drops out of the next day's run.
- Finance gets a daily overview for free — total reminders sent, total amount still outstanding, and per-client detail land in their inbox every morning without anyone compiling it.

3. Architecture Overview

The workflow is two independent flows sharing one Google Sheets ledger: a daily scheduled reminder cycle, and an always-on webhook that reacts the moment a payment is made.

Flow	Trigger	Purpose
Daily Reminder Cycle	Schedule trigger, once a day	Reads every unpaid invoice, decides who needs a reminder, drafts and sends it, and reports to finance
Payment Webhook	Webhook, called by a payment page	Marks the matching invoice as paid, emails a confirmation, and removes it from future reminder runs

Each unpaid invoice is placed into exactly one reminder tier per run, based on how many days it's overdue and how long since it was last reminded.

Timing	Reminder type	Urgency
3 days or fewer before due, no reminder sent yet	upcoming_reminder	Low
Due today, 1+ day since last reminder	due_today	Medium
Overdue, 7+ days since last reminder	first_reminder	Medium
Over 14 days overdue, 5+ days since last reminder	second_reminder	High
Over 30 days overdue, 3+ days since last reminder	final_notice	Critical

4. Daily Reminder Cycle — Step by Step

1

Run on schedule

A daily trigger starts the whole cycle at a configured time each morning.

2

Read every invoice

All rows are pulled from the Invoices tab in Google Sheets, covering every client and payment status on record.

3

Decide who needs a reminder

Paid invoices are skipped immediately. For everything else, days overdue and days since the last reminder are calculated, and each invoice is matched against the reminder tiers above — anything that doesn't yet qualify for a tier is simply left alone for today.

4

Build the AI prompt

For each invoice that needs a reminder, a structured prompt is assembled with the client name, invoice number, amount, dates, days overdue, reminder type and urgency level — with explicit instructions not to invent details or mention a payment link (the workflow adds that itself).

5

Draft the email with Groq

A Groq-powered agent writes a concise, professional reminder following a fixed structure (subject line, greeting, body, fixed sign-off), matching the tone to the urgency level without using placeholders or inventing company details.

6

Format the final email

The AI's draft is cleaned up, any stray placeholder text is stripped out, and the workflow wraps it in a styled HTML template with an invoice-details table and a Pay Invoice button — kept separate from the AI so the payment link is always correct and never hallucinated.

7

Send & update the ledger

The reminder email is sent to the client, then the invoice row is updated with the reminder timestamp, an incremented reminder count, and the reminder type that was just sent.

8

Log & report

Each send is recorded as a structured entry in a separate Activity Log tab, and once the day's run finishes, a summary — total reminders, total amount outstanding, urgency breakdown and a per-client table — is emailed to the finance team.

5. Payment Webhook — Step by Step

1

Receive the payment notification

A webhook accepts a POST request from the payment page, carrying the invoice number, amount and payment method.

2

Find the matching invoice

The Invoices sheet is searched by invoice number to locate the exact row to update.

3

Mark it paid

The row is updated with `payment_status` set to paid, along with the payment date, amount and method — so the next day's reminder run will skip it automatically.

4

Confirm both sides

The payment page receives a JSON success response, and the client is separately emailed a formatted payment confirmation with the invoice number, amount paid and method as a receipt.

6. Node-by-Node Map

Flow	Key nodes
Daily reminder cycle	Schedule Daily Check → Fetch Pending Invoices → Calculate Reminder Logic → Prepare AI Prompt → AI Agent For Generate Email Content (+ Groq Chat Model) → Format Email → Send Email Reminder → Update Reminder Status
Logging & reporting	Create Activity Log → Save to Activity Log → Generate Daily Summary → Send Summary to Finance Team
Payment webhook	Webhook: Payment Received → Lookup Invoice → Update Payment Status → Webhook Response / Send Payment Confirmation

7. Reliability & Security Design

- **Payment link never AI-generated** — the AI is explicitly instructed not to mention a payment link or button; the workflow builds that link itself from the invoice's real data, so it can never be wrong or hallucinated.
- **Placeholder stripping** — the email formatting step actively removes any stray [Your Name] / [Company Name]-style placeholders the AI might produce, so nothing unfinished reaches a client.
- **Rate-limited by design** — each reminder tier requires a minimum number of days since the last reminder, preventing a client from being emailed every single day by accident.
- **Payments stop reminders immediately** — the webhook updates payment_status the moment a payment lands, so there's no risk of a reminder going out after a client has already paid.
- **Full send history** — every reminder is both reflected on the invoice row (count, type, last sent) and recorded as its own entry in the Activity Log, giving a complete, queryable history per client.

8. Setup Checklist

- Connect a Google Sheets credential and point every Sheets node at your own spreadsheet, with an Invoices tab and an Activity Log tab.
- Connect a Groq credential for the AI Agent For Generate Email Content node.
- Replace the from-address placeholders (your-finance@example.com) in the reminder and payment-confirmation email nodes with your real sending address.
- Set the real payment page URL in Format Email so the Pay Invoice button links to your actual checkout, and update the finance team address in Send Summary to Finance Team.
- Activate the workflow, then point your payment page's success callback at the webhook's Production URL (path invoice-paid) so payments are recorded automatically.
- Adjust the reminder-tier thresholds (days overdue, days since last reminder) in Calculate Reminder Logic if your business needs a different cadence.

This workflow was designed and built end-to-end in n8n as part of an AI automation portfolio, demonstrating rule-based scheduling logic, AI-assisted email drafting with guardrails, and a webhook-driven payment reconciliation loop.